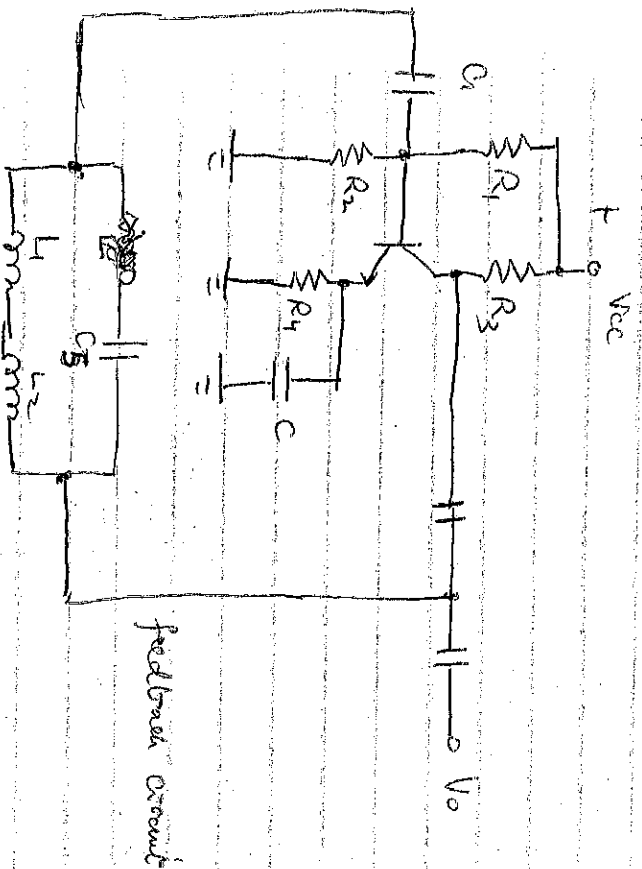


(a)

Oscillator

An oscillator is a circuit which gives an output without any input signal.
Wien bridge oscillator

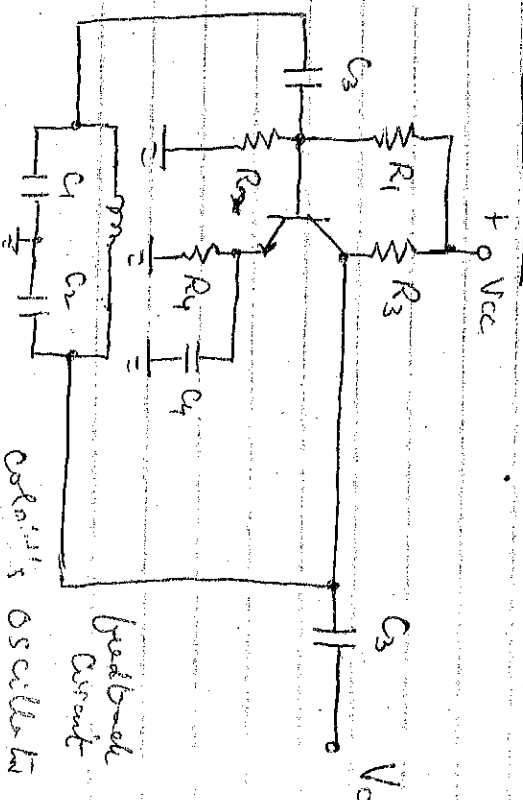


Wien bridge oscillator

$$f_r = \frac{1}{2\pi\sqrt{L_1 C_2}}$$

$$\text{where } L_1 = L_1 + L_2$$

$$\beta = \frac{k_1}{k_2} \text{ and } A = \frac{k_2}{k_1}$$

~~Cell~~Colpitts Oscillator

Colpitts Oscillator

(B2)

$$f = \frac{1}{2\pi \sqrt{LC}}$$

$$\text{where } C = \frac{C_1 C_2}{C_1 + C_2}$$

$$B = \frac{C_2}{C_1} \quad \text{and} \quad A = \frac{C_1}{C_2}$$

for sustainable oscillation the loop gain

$$AB = 1$$

phase 5π of oscillators give sinusoidal wave form

Multivibrator

There are three types of multivibrators called Astable, bistable and monostable. These circuits give square waves unlike Hartley or Colpitts oscillator.

Astable This is the circuit that oscillates between two states, HIGH and LOW without any external trigger.

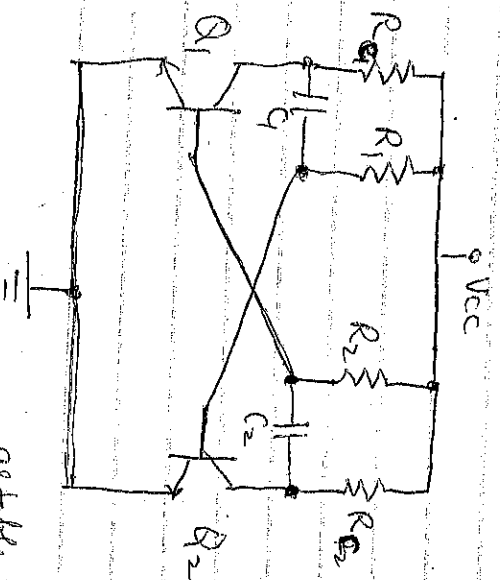
Bistable: This is the circuit that may be in any one of the two states indefinitely. It can only switch state to another stable state given an external trigger.

Monostable is the one which is stable in only one state.

If triggered it goes into unstable state briefly depending upon the timing components before reverting to the stable state.

Astable

Astable multivibrator: Derivation of frequency of operation of the circuit



astable multivibrator circuit

(3)

If $T_1 = 0.69 R_1 C_1$ high state time constant
 $T_2 = 0.69 R_2 C_2$ low state time constant
then $T = T_1 + T_2 =$ time period of the circuit

For a 50% duty cycle = $\frac{T_2}{T_1 + T_2}$

$$\frac{50}{100} = \frac{T_2}{T_1 + T_2}$$

$$100T_2 = 50T_1 + 50T_2$$

$$\therefore 50T_2 = 50T_1$$

$$T_2 = T_1$$

$$\text{for } T_2 = T_1, \quad R_1 = R_2 = R$$

$$C_1 = C_2 = C$$

$$\therefore T = T_1 + T_2 = 0.69RC + 0.69RC \\ = 1.38RC$$

$$\text{but } f = \frac{1}{T} = \frac{1}{1.38RC}$$

To choose the frequency, keeping the same square-wave space, both RC time constant will be changed together, at the same time. But to change both frequency and mark-space ratio, only one RC time constant network need to be changed.

Applications of multivibrator in digital circuits.

- for timing digital circuits (clock)
- for frequency digital circuits
- for synchronization of digital circuits

Example 1. Design an astable multivibrator using transistors given the following specifications.

frequency of operation = 50 kHz

$$V_{BE1} = 0.3V \text{ for transistor } Q_1$$

$$V_{BE2} = 0.3V \text{ for transistor } Q_2$$

$$V_{CE1} = 0.3V \text{ for transistor } Q_1$$

$$V_{CE2} = 0.3V \text{ for transistor } Q_2$$

$$\beta_1 \approx 100 \text{ for transistor } Q_1$$

$$\beta_2 = 100 \text{ for transistor } Q_2$$

(4)

$$V_{cc} = 12V$$

$$I_{C1} = 20mA \text{ for } Q_1$$

$$I_{C2} = 20mA \text{ for } Q_2$$

Let R_{C1} and R_{C2} be collector resistance for Q_1 and Q_2

$$\therefore V_{CE1} = V_{cc} - V_{CE1}$$

$$= 12 - 0.3 = 11.7V$$

$$\therefore R_{C1} = \frac{V_{CE1}}{I_{C1}} = \frac{11.7}{20mA} = 585\Omega$$

$$\text{Similarly for } R_{C2} = \frac{V_{CE2}}{I_{C2}} = \frac{11.7}{20mA} = 585\Omega$$

$$\text{For 50\% duty cycle} = \frac{T_2}{T_1 + T_2}$$

$$\frac{50}{100} = \frac{T_2}{T_1 + T_2}$$

$$50T_1 + 50T_2 = 100T_2$$

$$50T_1 = 50T_2$$

$$T_1 = T_2$$

$$\therefore R_{C1} = R_{C2} = R_C$$

$$C_1 = C_2 = C$$

$$\text{Min base Current } I_{Bmin} = \frac{I_C}{\beta}$$

$$\text{but s.t. } I_B = 3 \times I_{Bmin} = 3 \times \frac{I_C}{\beta}$$

$$\therefore R_{B1} = \frac{V_{cc} - V_{BE1}}{I_{B1}}$$

$$\text{but } I_{B1} = 3 \times \frac{I_{C1}}{\beta_1} = \frac{3 \times 20mA}{100} = 600\mu A$$

$$\therefore R_{B1} = \frac{12 - 0.3}{600\mu A} = 19.5k\Omega$$

$$\text{Similarly } R_{B2} = 19.5k\Omega$$

$$\text{from } f = \frac{1}{1.38RC}, \quad C = \frac{1}{1.38 \times 19.5k \times 50kHz}$$

$$= 0.7\mu F$$

(5)

